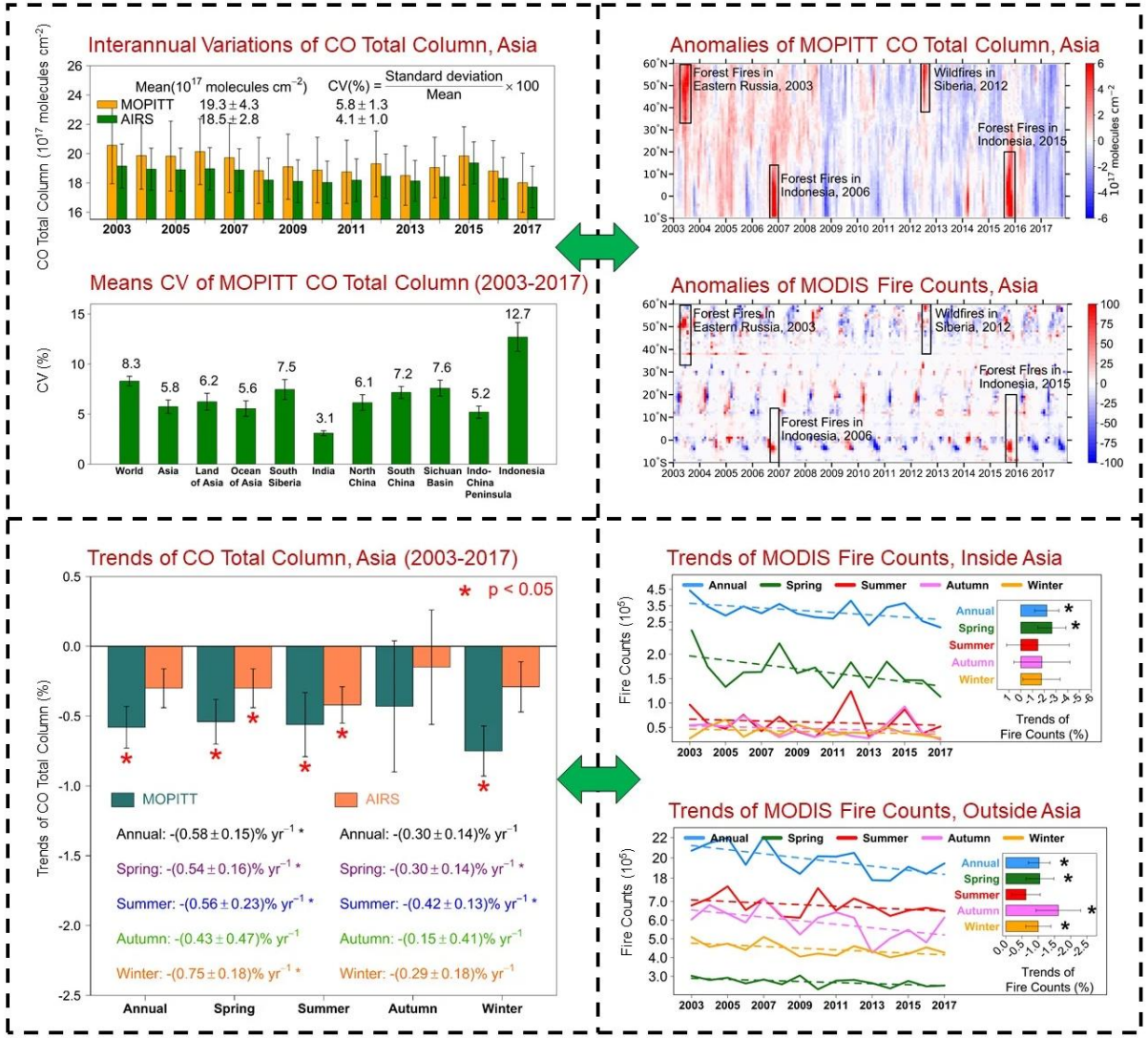
Research Summary

1. Satellite-Observed Variations and Trends in Carbon Monoxide over Asia and Their Sensitivities to Biomass Burning



1. The interannual variation of the CO total column in Asia is significantly and strongly correlated with biomass burning. The total column of CO in Asia presents significant inter-annual variability, as demonstrated by a coefficient of variation (CV) of 5.8% in the MOPITT CO total column data.

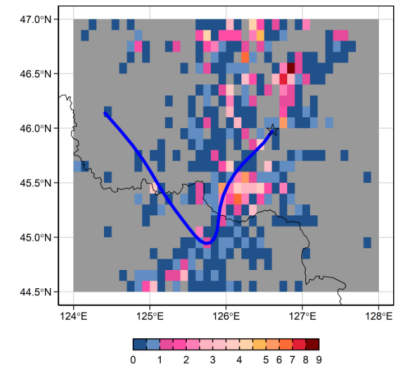
2. The trend in total CO columns over Asia aligns with the declining trends in fire counts and CO emissions from fires, both within and outside Asia. From 2003 to 2017, the MOPITT CO total column in Asia decreased at a rate of -0.58±0.15% per year. From 2003 to 2017, the decreasing trend of the total number of MODIS fire count within and outside Asia decreased significantly, suggesting that biomass burning is one of the reasons for the decreasing trend of CO total column in Asia, especially in spring.

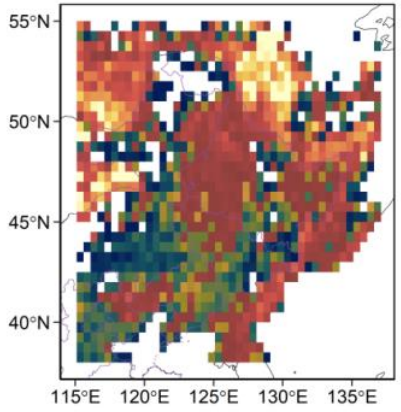
***Zhang et al.* (2020)**

Research Summary

2. Impact of strong biomass burning in Northeast China: air pollution, correlations among pollutants, and health risk

Biomass Burning in Northeast China (2014-2020)
18 cities, 121 sites





841 fire-affected days
Cropland fires: 75%

HYSPLIT+MODIS

Transport

Impact on Air Quality

$PM_{2.5}$	+38.7 μg/m³	↑106%
CO	+105.1 ppb	↑20%
NO_2	+4.1 ppb	↑36%
O_3	+6.4 ppb	↑18%

Ozone Photochemistry Production
Fire plumes (spring & fall)

O_3 Production in Fire Plumes

$NO_x + VOCs + CO + h\nu \rightarrow O_3$

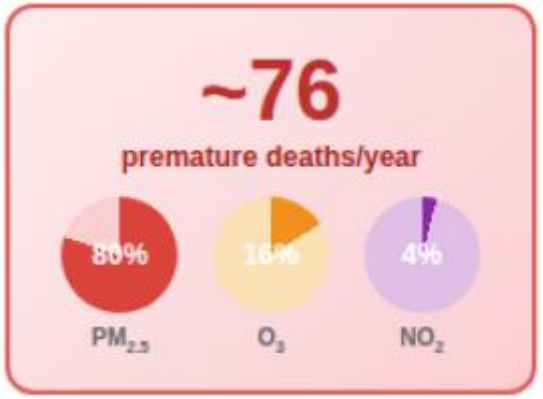
$\Delta O_3/\Delta CO$	$\Delta O_3/\Delta NO_2$
0.050 ppb/ppb	2.40 ppb/ppb

Exposure

Exposure to Cropland Fires
$PM_{2.5}$, O_3, and NO_2

Health

Short-term All-cause Premature Deaths



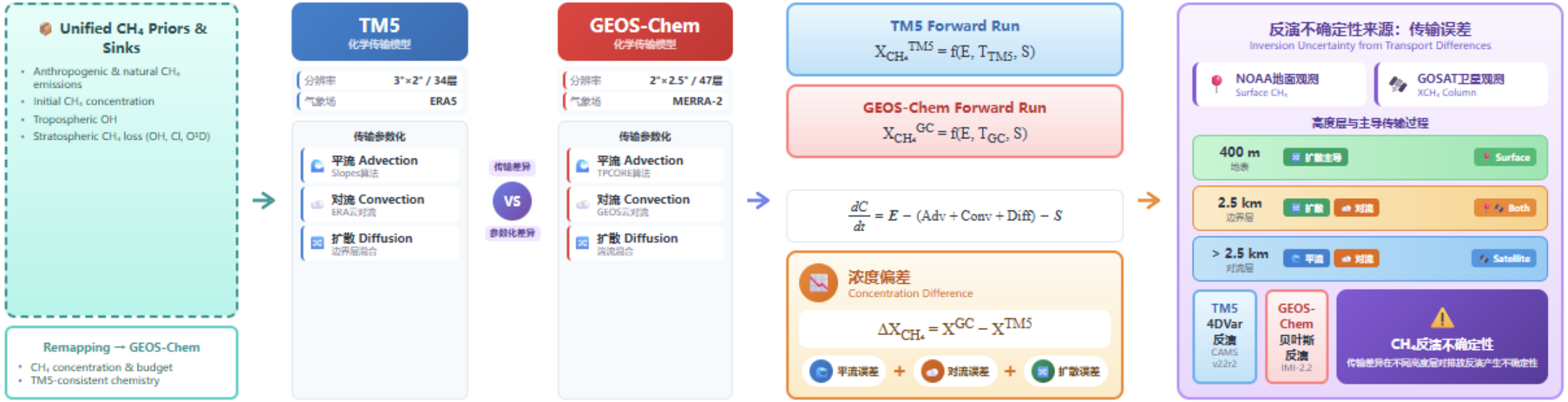
Spring: O_3 = 27% mortality

1. Fall enhancements in air pollutants 2-3x than spring
2. O_3 enhanced only in spring
3. Low $\Delta O_3/\Delta CO$ ratio in cropland fires vs tropical fires

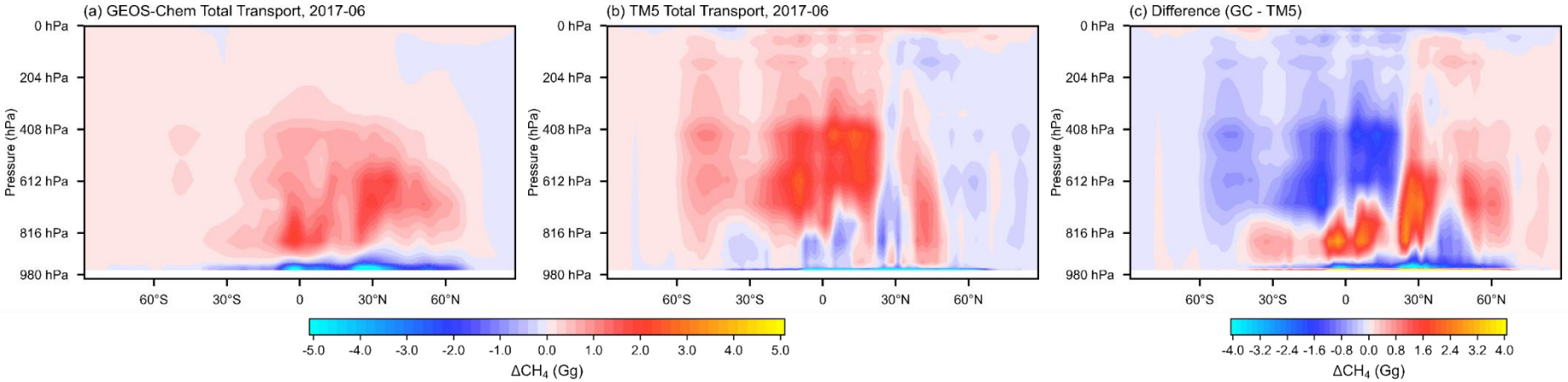
Zhang et al. (2026) Under review **AE**

Research Summary

3. **Quantifying transport-induced uncertainties in global methane inversions: a multi-model evaluation using GEOS-Chem and TM5 constrained by surface and satellite observations.**



Same CH_4 source → Concentration bias → Inversion bias



Zhang et al. (2026)